

Nafis Ahtasum

MATERIALS & CHEMICAL RESEARCHER · EXPERIMENTAL BENCHMARK · MACHINE LEARNING · FIRST-PRINCIPLES DFT
Center for Material, Climate and Energy (CMCE), RAI · 116/A, Tejgaon Industrial Area, Dhaka-1208, Bangladesh

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RESEARCH INTERESTS

Closed-Loop Materials Design & Interfacial Engineering: Bridging **experimental benchmark** (hands-on synthesis, spectroscopy, particle kinematics, and **multifunctional polymer composite processing**) with **atomistic modeling (first-principles DFT/MD)** and **physics-informed generative AI** (CDVAE, GNNs) to interrogate **defects in crystalline materials**, break catalytic scaling relations in selective CO₂ reduction (CO₂RR), suppress parasitic HER, and design high-performance macromolecular and energy systems.

EDUCATION

Bangladesh University of Textiles (BUTEX)

Dhaka, Bangladesh

B.Sc. in Textile Engineering (Specialization: Materials & Polymer Engineering)

Graduation: Aug. 2026

- **Undergraduate Experimental Thesis:** *Comparative Analysis of Microfiber (Microplastic) Release from Industrial Washing Techniques Using Machine Learning and Optical Image Processing.*
- **Core Laboratory & Engineering Coursework:** Polymer Science & Engineering, Physical Chemistry, Chemical Reaction Dynamics, Fiber Physics, Computational Modeling, Engineering Mathematics, Material Synthesis, and characterization.

PUBLICATIONS (Full list on Google Scholar)

- **Ahtasum, N.,** Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., and Roni, M. R. A., 2026. **Backward mapping from target optoelectronic phenotypes to chemical genomes for the discovery of lead-free double perovskites.** *Materials Today Physics* (Q1, IF: 9.3), 66, p.102180. [🔗 GitHub Code & Pipeline](#)
- **Ahtasum, N.,** Sohan, S. R., Himel, M. M. A., Hassan, M. Z., Moklis, M. H., Rashel, M. R., Jamil, H., Islam, A. K., and Tlemcani, M., 2026. **Genome-Guided Interpretable Screening of Phase-Stable, Lead-Free Double Perovskite Absorbers for All-Inorganic Semiconductors, Sensors, and Photovoltaics with DFT-Validated Design Rules.** *arXiv Preprint*, arXiv:2605.22887. [🔗 GitHub Code & Pipeline](#)

RESEARCH EXPERIENCE

Bangladesh University of Textiles (BUTEX)

Dhaka, Bangladesh

Undergraduate Researcher (Supervisors: Most. Jesmin Nahar Kamona, Nusrat Jahan)

Jan. 2024 – May 2025

- **Polymer Composite Processing & Mechanical Testing:** Formulated compression-molded composite panels blending textile polymer offcuts into polypropylene (PP) matrices via a CARVER hydraulic hot press; conducted mechanical UTM tensile and flexural testing outperforming ASTM D350 standards (*JFPC*, 2024). [🔗 Published Paper](#)
- **Microscale Environmental Tracking & Characterization:** Executed multi-stage membrane filtration, gravimetric degradation tracking, and surface morphology tracking via FE-SEM/EDS, FTIR-ATR, and ImageJ optical particle kinematics; presented at the *Intl. Conference on Textile Science and Engineering 2025*.

Center for Material, Climate and Energy (CMCE), RAI

Dhaka, Bangladesh

Materials Researcher (Supervisor: Dr. M. M. Harussani)

2024 – Present

- **Generative Inverse Design & Defect Engineering for CO₂RR Electrocatalysis** **Ongoing**
Developing a property-to-structure generative framework coupling CDVAE, Swarm Intelligence (BSA), and directional GNNs (DimeNet++) to design defect-engineered alloy catalysts:
 - **Defect-Graph Encoding:** Formulated crystal graphs encoding surface point vacancies, undercoordinated sites, and dopants; trained directional GNNs to predict intermediate binding energies over perturbed local coordination environments.
 - **Microkinetic Decoupling:** Guided swarm optimization to break $\ast\text{COOH}$ vs. $\ast\text{CO}$ linear scaling, optimizing volcano-apex $\ast\text{CO}$ binding ($|\Delta E_{\text{CO}} + 0.67 \text{ eV}| \leq 0.15 \text{ eV}$) while penalizing parasitic HER ($\Delta E_{\text{H}} + 0.27 \text{ eV} \geq 0.6 \text{ eV}$).
 - **DFT Thermodynamic Screening:** Automated Pymatgen/VASP workflows to evaluate defect formation energies (E_{form}), defect interaction landscapes, and ZPE corrections, isolating stable defective catalytic states.
- **High-Throughput Material Discovery Pipeline:** Built an evolutionary-tuned ML screening pipeline across 13,088 $\text{A}_2\text{BB}'\text{X}_6$ compositions ($R^2 = 0.93$). Formulated a high-recall loss function (0.83 recall) to retain viable metastable phases ($\Delta E_{\text{hull}} \leq 25 \text{ meV/atom}$). Applied SHAP feature importance to narrow search space by 99.9%, isolating 7 DFT-validated stable candidates with 90% reduced excitonic drag for $\text{Cs}_2\text{SnGeBr}_6$ ($\mu^\ast = 0.045 m_0$, $\varepsilon_1(0) = 7.54$). [🔗 GitHub Repo](#)

- **Double Perovskite Oxide Screening Pipeline:** Developed an independent multi-objective machine learning pipeline trained on $> 5,000$ double perovskite oxide ($A_2BB'O_6$) compositions to evaluate thermodynamic stability, classification of metal/insulator transitions, and optimal bandgap target windows. [GitHub Repo](#)
- **Stimuli-Responsive Nanomaterials & Adsorption Dynamics:** Simulated adsorption mechanisms and pH-dependent release kinetics of capecitabine on Ti_3C_2 MXene surfaces using DFT and MD; quantified strong binding ($E_{ads} \approx -3.2$ eV) stabilized by $0.3\text{--}0.4|e|$ charge transfer and calculated tumor-acidic release pathways ($\Delta E_{rel} \approx +3.2$ eV). [Manuscript in Prep](#)
- **Electrocatalytic Catalyst Screening:** Executed high-throughput DFT calculations on $> 400k$ candidate structures generated by a deep generative framework, identifying 18 HER-suppressing Co-Zr-X ($X = Ba, Ti$) trimetallics for cathodic glycerol electroreduction. [Manuscript in Prep](#)

Department of Computer Science, University of Idaho

Research Collaborator (Supervisor: Prof. Hasan Jamil)

Remote, USA

June 2026 – Present

- **ML Surrogate-Driven Inverse-Design Pipeline Development:** Developing inverse-design computational workflows, curating benchmark materials datasets, and training ML surrogate models to screen phase-stable perovskite systems for targeted optoelectronic applications. [Ongoing](#)
- **Genome-Guided Physics-Informed ML:** Developed 3 descriptor families across 13 structural variables with staged constraint filtering over 13,000 compositions, isolating 5 DFT-confirmed strictly stable semiconductors ($E_{hull} \leq 0$ meV/atom).

[GitHub Repo](#)

Universidade de Évora

Research Collaborator (Supervisors: Prof. Dr. Mouhaydine Tlemcani, Dr. Masud Rana Rashel)

Remote, Portugal

Jan. 2025 – Apr. 2026

- Formulated evolutionary-tuned ML pipelines and materials feature databases to screen phase-stable halide double perovskites, coupling surrogate model predictions with DFT validation. [Manuscript in Prep](#)

TECHNICAL, EXPERIMENTAL & COMPUTATIONAL SKILLS

Experimental Characterization:	Field-Emission Scanning Electron Microscopy & Energy-Dispersive Spectroscopy (FE-SEM/EDS), Fourier-Transform Infrared Spectroscopy (FTIR-ATR), automated ImageJ particle kinematics/tracking, Universal Testing Machines (UTM tensile/flexural evaluation), optical microscopy.
Synthesis & Materials Processing:	Compression molding via CARVER hydraulic hot presses , thermoplastic polymer matrix compounding (Polypropylene/PP), multi-stage membrane filtration, chemical treatment baths, gravimetric degradation analysis, ASTM standard mechanical testing.
DFT, MD & Electronic Structure:	Density Functional Theory (DFT) and Molecular Dynamics (MD) using CASTEP, Quantum ESPRESSO, DMol3, VASP, Materials Studio, Gaussian 09W, and VESTA; automated Pymatgen slab construction, charge transfer, TDOS/PDOS, band structures.
Machine Learning & Informatics:	Crystal Diffusion Variational Autoencoders (CDVAE), Directional/Equivariant GNNs (DimeNet++ , GemNet, MEGNet), Swarm Intelligence (Bird Swarm Algorithm), evolutionary optimization (DEAP), high-recall tree ensembles, SHAP; PyTorch Geometric, Scikit-learn.
Catalysis & Defect Engineering:	Point-defect thermodynamics (E_{form}), surface vacancies/dopants, microkinetic scaling relations, Computational Hydrogen Electrode (CHE) model, intermediate adsorption free energies (ΔG_{*CO} , ΔG_{*H} , ΔG_{*COOH}), competitive HER suppression.
Software & HPC Workflow:	Linux/Bash cluster environments, Git/GitHub version control, VS Code, LaTeX, OriginPro, Adobe Illustrator.

LEADERSHIP, MENTORSHIP & OPEN SCIENCE

Resogram (Founder & Product Lead)

2025 – Present

Architected an open academic network connecting **1,001+ active global researchers**, leading system design for interdisciplinary research matching, automated profile indexing, and transparent pre-publication feedback.

Research and Analysis Institute (RAI) (Founder & Managing Director)

2024 – Present

Direct computational training initiatives, providing students with open-source curriculum and HPC resources. Instructed 4 technical workshops including *7-Day Hands-on DFT Simulation Training* (56 attendees) and *Quantum ESPRESSO & Materials Studio Bootcamp* (122 attendees). Co-supervised 2 student DFT projects on defect kinetics in $Cs_2AgBiCl_6$ and H_2SO_4 adsorption on MXenes.

V-Trion Textile Research GmbH (Lead Research Fellow, Supervisor: Dr. Ashaduzzaman Khan)

July 2025 – Present

Direct 6 remote research teams (28 students) conducting systematic reviews on solid polymer electrolytes, wearable elec-

trochemical biosensors, silicon-anode binders, and direct air capture (DAC) sorbents.

Science Policy & Systems Researcher, PIRI-BD Initiative

Apr. 2025 – Present **Manuscript in Prep**

Supervisors: Md. Rafiul Alam Roni and Prof. A. K. M. Kamrul Islam

Dhaka, Bangladesh

- Co-authoring the strategic vision manuscript "*Towards the Establishment of a Private Interdisciplinary Research Institute*" alongside international faculty (NC A&T, Univ. of Iowa, Univ. of Maine, Univ. of Évora), formulating operational blueprints for research governance and funding.

PROFESSIONAL EXPERIENCE

Managing Director, Research and Analysis Institute (RAI), Dhaka, Bangladesh

2026 – Present

Computational Materials Researcher, Center for Material, Climate and Energy (CMCE), RAI

2024 – Present

Research Assistant, Universa Pulsar Lda (Spin-off of Universidade de Évora), Évora, Portugal

2025 – 2026

Industrial Trainee, Knit Concern Ltd., Narayanganj, Bangladesh

2025

ACADEMIC REFERENCES

Most. Jesmin Nahar Kamona

Assistant Professor, Apparel Eng.

Bangladesh University of Textiles

✉ jasmine@butex.edu.bd

Prof. Hasan Jamil, Ph.D.

Associate Professor, Computer Science

University of Idaho, USA

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Dr. M. M. Harussani, Ph.D.

Principal Investigator, CMCE (RAI)

Postdoc Fellow, Universiti Malaya

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